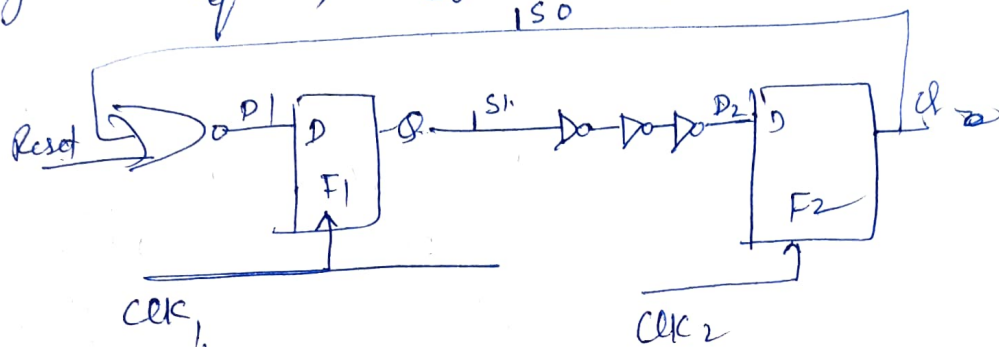


Advd

$t_{cd} = 1ns$ $t_{pd, logic} = 2ns$ inverter

$t_{cd} = 2ns$ $t_{pd, logic} = 3ns$ NOR

D register $t_{cd} = 1ns$, $t_{su} = 2ns$ $t_h = 2.5ns$



a) Clk period T_{clk} .

Path. $F_1 \rightarrow F_2$:-

$$T_{clk} \geq T_{cq, F1} + T_{pd, logic} + T_{su, F2}$$

(3 inv.)

$$\geq 1ns + 6ns + 2ns$$

$$T_{clk} \geq 9ns$$

$t_{hold} < T_{cq} + t_{cd, logic}$
 $< 1ns + 3ns$
 $t_h < 4ns$
 $2.5 < 3ns$
 valid.

Path $F_2 \rightarrow F_1$:-

$$T_{clk} \geq T_{cq, F2} + T_{pd, logic} + T_{su, F1}$$

$t_{hold} < T_{cq, F2} + t_{cd, logic}$
 $t_h < 1ns + 2ns$
 $2.5 < 3ns$
 valid.

So $T_{clk} \geq 9ns$

smallest clk period
 $freq = \frac{1}{9} ns = 83.3MHz$

(b) When 2 inverters are removed

Path $F_1 \rightarrow F_2$:- $t_{hold} < 1ns + 1ns$
 $2.5 \not< 2ns$ not valid.

Path $F_2 \rightarrow F_1$:- $t_{hold} < 1ns + 2ns$
 $2.5 < 3ns$ valid.

hold time violation
 circuit will not function

(a) Min time period of clock:- $\text{Skew} \pm 1.5\text{ns}$

Path $F_1 F_2$: $T_{\text{cell}} \geq T_{\text{eqP}} + T_{\text{logic}} + T_{\text{setup}} + t_{\text{skew}}$

$T_{\text{cell}} \geq 1\text{ns} + 6\text{ns} + 2\text{ns} \mp 1.5\text{ns}$

≥ 1 7.5ns (+ve skew)

≥ 1 10.5ns (-ve skew)

Path $F_2 F_1$: $T_{\text{cell}} \geq T_{\text{eqP}_2} + T_{\text{logic}_{\text{NOR}}} + T_{\text{setup}}$

$\geq 1\text{ns} + 3\text{ns} + 2\text{ns} \mp t_{\text{skew}}$

≥ 1 $6 \mp 1.5\text{ns}$

≥ 1 7.5ns (+ve skew)

≥ 1 4.5ns (-ve skew)

$t_{\text{hold}} < t_{\text{eq}} + t_{\text{cd}}$
 $< 1\text{ns} + 3\text{ns} - 1.5\text{ns}$
 $< 2.5\text{ns}$

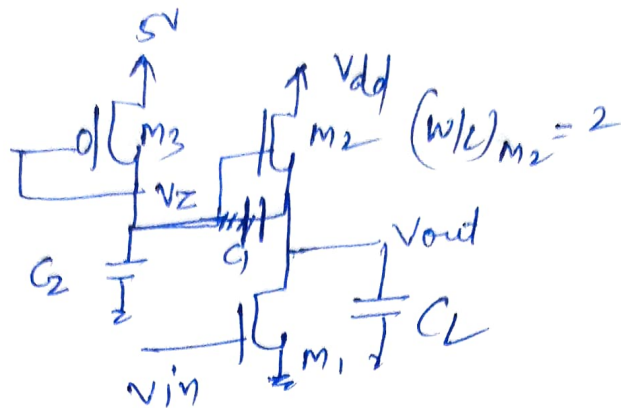
$t_{\text{hd}} < t_{\text{eq}} + t_{\text{cd}} - t_{\text{logic}}$
 $< 1\text{ns} + 2\text{ns} - 1.5\text{ns}$
 $t_{\text{hd}} < 1.5\text{ns}$

So for all paths to operate

$T_{\text{cell}} \geq 10.5\text{ns}$
 $t_{\text{h}} \leq 1.5\text{ns}$

2

Q.2



(a) $|V_T| = 0.8V$
 $\mu_n C_{ox} = 102 \mu A/V^2$
 $(w/L)_2 = 2, (w/L)_{M1} = 20$
 $V_{DD} = 5V$

(a) $V_Z = V_{DD} - |V_{TP}| = 5 - 0.8 = 4.2V$ (2)
 Neglecting body effect.

$V_{OH} = (V_{DD} - |V_{TP}|) - |V_{TN}| = 4.2V - 0.8V = 3.4V$

$V_{OL} \Rightarrow M_2 \text{ in sat, } M_1 \text{ in lin.}$

$$\frac{\mu_n C_{ox}}{2} (w/L)_2 (4.2V - V_{OL})^2 = \frac{\mu_n C_{ox}}{2} (w/L)_{M1} \left[2(V_{OH} - V_T)V_{OL} - V_{OL}^2 \right]$$

$$1 \times [4.2 - V_{OL}]^2 = 10 [2(3.4 - 0.8)V_{OL} - V_{OL}^2]$$

$$\Rightarrow V_{OL}^2 - 8.4V_{OL} + 17.64 = 10 [2 \times (2.6)V_{OL} - V_{OL}^2]$$

$$\Rightarrow V_{OL}^2 - 8.4V_{OL} + 17.64 = 10 [5.2V_{OL} - V_{OL}^2]$$

$$\Rightarrow 11V_{OL}^2 - (8.4 + 52)V_{OL} + 17.64 = 0$$

$$11V_{OL}^2 - 60.4V_{OL} + 17.64 = 0$$

$$11V_{OL}^2 - 60.4V_{OL} + 17.64 = 0 \Rightarrow V_{OL} = 0.31V$$
 (3)

$V_m \Rightarrow M_1, M_2 \text{ in saturation.}$

$$1 [4.2 - V_m]^2 = 10 [V_m - 0.8]^2$$

$$17.64 + V_m^2 - 8.4V_m = 10 [V_m^2 - 1.6V_m + 0.64]$$

$$9V_m^2 - 10.0V_m - 18.28 = 0$$

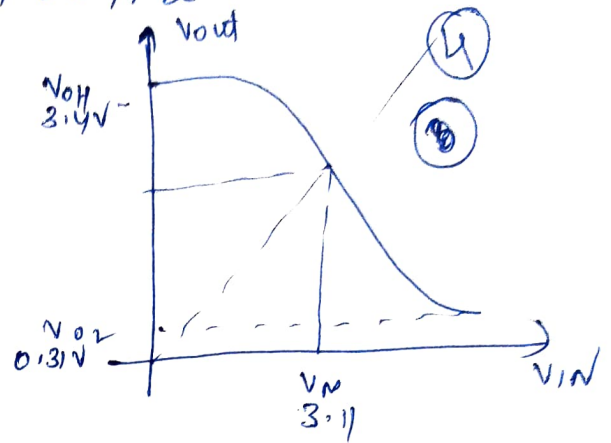
$$17.64 + V_m^2 - 8.4 V_m = 10 V_m^2 - 16 V_m + 6.4$$

$$9 V_m^2 - (16 + 8.4) V_m + (6.4 - 17.64) = 0$$

$$9 V_m^2 - 24.4 V_m - 11.24 = 0$$

$$V_m = 3.11 V$$

(3)



(b) Ratio (C_2/C_1) for full logic o/p

$$\Rightarrow \frac{V_{DD}}{C_1} = V_Z$$

after

$$V_Z = \frac{C_1}{C_1 + C_2} V_{DD} = \left(1 + \frac{C_1}{C_2}\right) \times 5V$$

$$(5.8) = \left(1 + \frac{C_1}{C_2}\right) 5V$$

$$\Rightarrow \frac{C_1}{C_2} = \frac{5.8}{5} - 1 = 0.16$$

$$\frac{C_2}{C_1} = 6.25$$

(4)

$$\Rightarrow \frac{C_2}{C_1} = 6.25$$

(c) if M_2 is made PMOS $\rightarrow V_Z = \text{Either } V_{DD} / V_{DD} + V_T$

$$\left. \begin{array}{l} V_{GS} = 0 \\ V_{GS} = -V_T \end{array} \right\} M_2 \text{ is off.}$$

Ckt. will not operate (5)

If $V_{DD} - V_T \rightarrow$ PMOS Weak
 V_{OH} may not reach fully

Ques 3
2

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

As electrical parameters remain same after use of high- ϵ gate dielectric.

$$\frac{\epsilon_{SiO_2}}{t_{SiO_2}} = \frac{\epsilon_{HfO_2}}{t_{HfO_2}}$$

$$t_{SiO_2} = \frac{\epsilon_{HfO_2}}{\epsilon_{SiO_2}} \times t_{HfO_2} = \frac{3.9}{25} \times 10 = 1.56 \text{ nm}$$

for any medium,

$$D_{SiO_2} = D_{HfO_2}$$

$$\epsilon_{SiO_2} E_{SiO_2} = \epsilon_{HfO_2} E_{HfO_2}$$

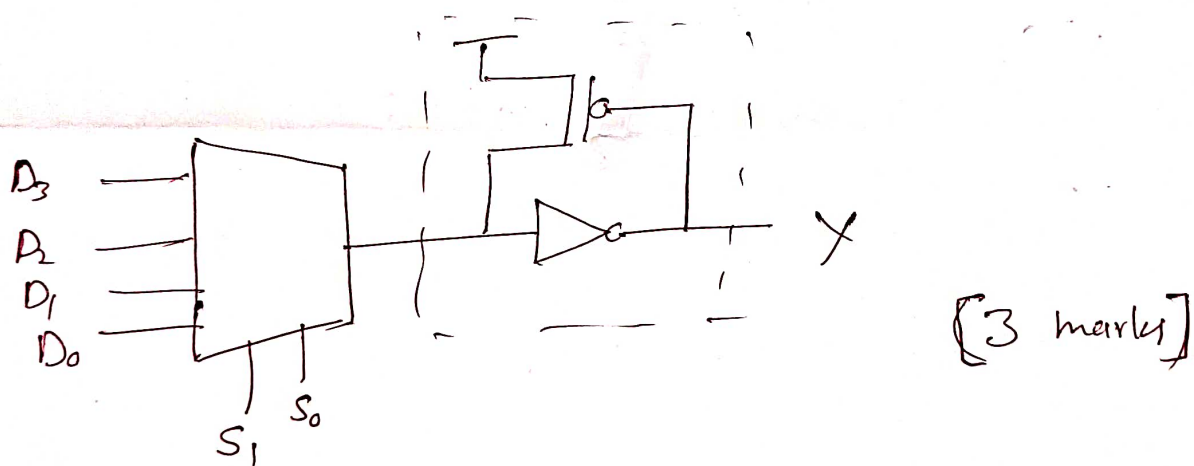
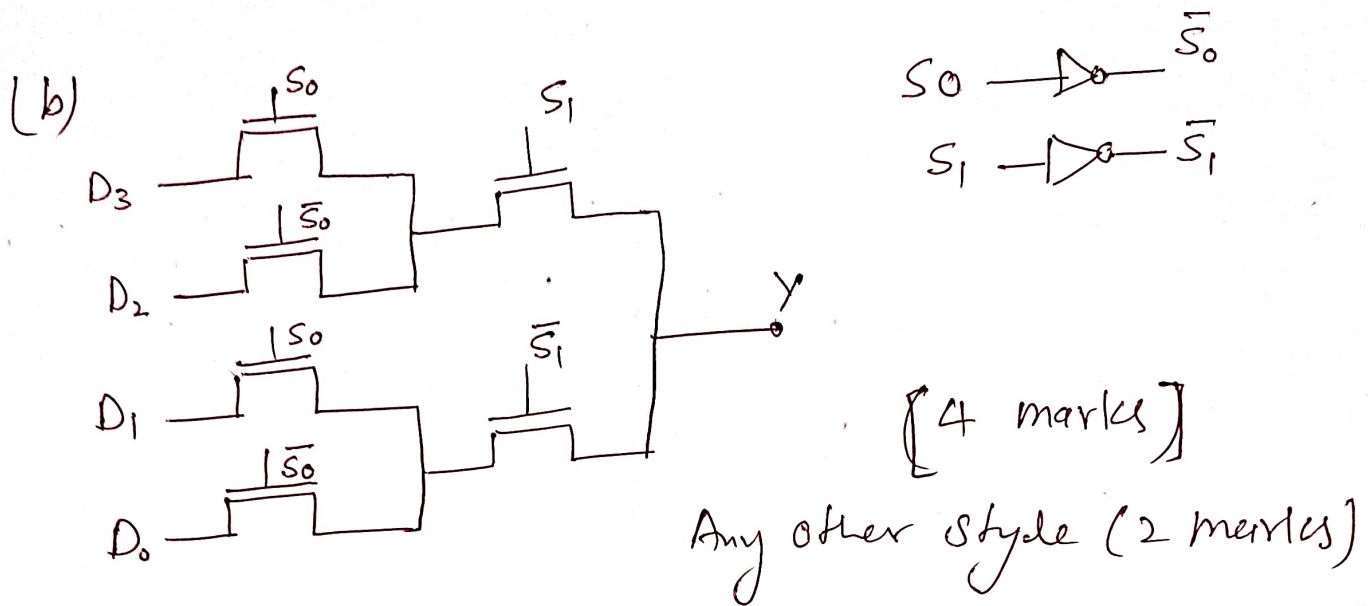
$$E_{HfO_2} = \frac{\epsilon_{SiO_2} E_{SiO_2}}{\epsilon_{HfO_2}} = \frac{3.9}{25} \times 25 \times 10^3 = 3.9 \times 10^3 \text{ V/cm}$$

$$E_{HfO_2} = 3.9 \times 10^3 \text{ V/cm}$$

Que 3 Solution (a)

$$d_{HfO_2} = 1.56 \text{ nm} \quad [2.5 \text{ marks}]$$

$$E_{HfO_2} = 3.9 \times 10^3 \text{ V/cm} \quad [2.5 \text{ marks}]$$



PMOS - weak PMOS (slow u/L) [1 mark]

Inverter - The output from PTL has a degraded logic '1' ($V_{DD} - V_{TO}$) and a perfect logic '0' (0V)

$\therefore$ we need a low skew inverter

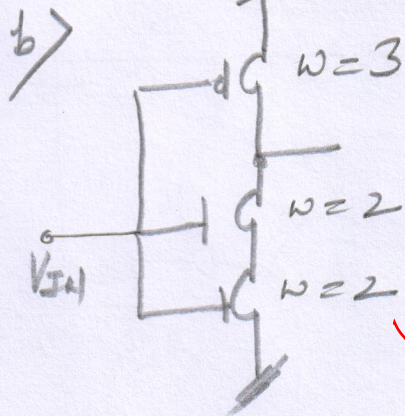
$$(P_p < P_n)$$

[2 marks]

a) Inverter

(4M)

Given: $\mu_n = 3 \mu_p$
 $f = 100 \text{ MHz}$
 $C = 1 \text{ pF}$



(2M)

sizing

(2M)

$V_{TN} = V_{TP}$
 $\mu_n = 3 \mu_p$ Given

$$g_{VIN} = \frac{3+2+2}{3 \cdot 4} = \frac{7}{3 \cdot 4}$$

$$g_{VIN} = \frac{6+6+1}{3 \cdot 4} = \frac{13}{3 \cdot 4}$$

c)

$$P_{sc} = \frac{K}{12} (V_{DD} - 2V_t)^3 \frac{t_{n,f}}{t_{p,in}}$$

$k_n = k_p = K$

$t_{p,in} = \text{period of I/P}$
 one cycle

$t_{n,f} = \text{rise/fall time}$

$$= \frac{120 \times 10^{-6}}{12} (2.5 - 0.8)^3 \times \frac{40 \times 10^{-12}}{10 \times 10^{-9}}$$

$$P_{sc} = 10 \times 10^{-6} (1.7)^3 \times 10^{-3} = 49.13 \times 10^{-9}$$

$P_{sc} = 49.13 \text{ nW}$

(4M)

d)

$$P = C_L V_{DD}^2 f \quad (\text{Neglect load}) \therefore P = V^2 f$$

Both of the circuit is operating at same freq & same voltage therefore they have nearly equal dynamic power consumption.

(2M)

e) static ckt power consumption
in fig 3a & fig 3b is same
but fig 3a has lower delay
due to lower value of $g = 7/3$

Therefore, circuit in fig 3a is
better in terms of lower & delay product.

Propagation delay
of circuit 3(b)
is large due to
large valued logical effort
 $g = 13/3$

└ (2M)

Q1) $|V_{Top}| = 0.8V$

$$I_{out} = 5I_{Ref} = 5 \times 20\mu A = 100\mu A$$

$$100\mu A = 0.1mA$$

wide range of voltage

$$(80\% \text{ of } V_{DD}) = 2.4V$$

$$\Rightarrow 2V_{OV} = 0.6V$$

$$V_{OV} = \underline{0.3V}$$

$$M1, M2 \Rightarrow \frac{W}{L} = 2.2$$

$$g_m = \frac{2I_D}{V_{OV}} = 667\mu A/V$$

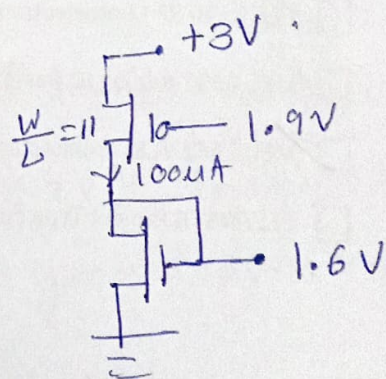
$$M3, M4 \Rightarrow \frac{W}{L} = (2.2) \times 5 = 11$$

$$r_{o3,4} = \frac{1}{\lambda I_{out}} = 1M\Omega$$

$$R_{out} = g_m r_o + 2r_o = 668M\Omega (> 200M\Omega)$$

Power Consumption:-

$$(100\mu A + 100\mu A + 20\mu A) \times 3 = 660\mu A = \underline{0.66mW}$$

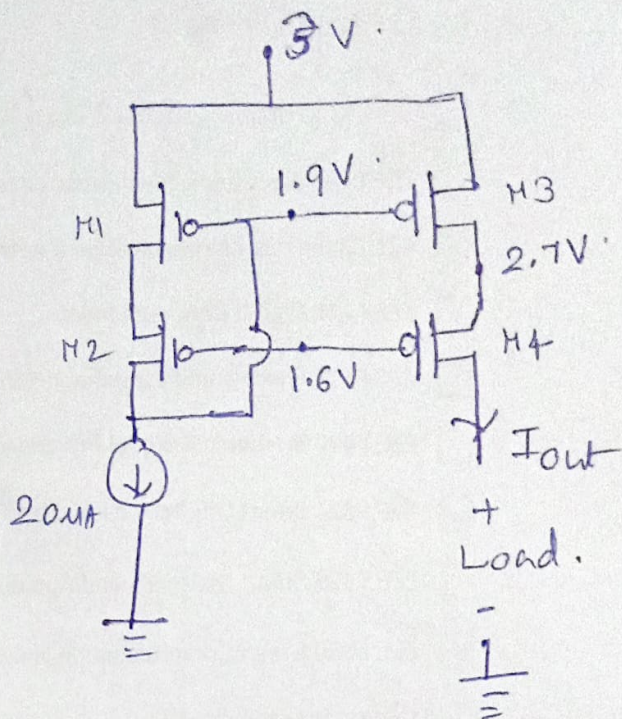


$$\frac{1}{2} \times 0.6mA/V^2 \times 0.8 \times \frac{W}{L} = 100\mu A$$

$$\left(\frac{W}{L}\right) = 0.52$$

Any other solution shall be considered with some penalty.

* "SOURCE" :- Use PMOS Current mirror



* If someone used $V_{EB1} = 0.6$ & calculated V_{EB2} NO marks deducted.

(Q2):-

$$V_{Ref} = V_{out} = V_{EB2} + \left(1 + \frac{R_2}{R_3}\right) V_T \ln(n)$$

(a) $1.2V = 0.6V + \left(1 + \frac{10K}{R_3}\right) 26mV \ln(15)$

$$R_3 = \frac{10K}{1.52} = \underline{1.329 K\Omega}$$

(b) $\frac{\partial V_{Ref}}{\partial T_{emp}}$ is now proportional to $\left(1 + \frac{R_2}{R_3}\right) \frac{KT}{q} \ln(15)$

$$\left(1 + \frac{10K}{R_3}\right) 26mV \ln(15) \approx 40mV$$

$\therefore$ for no value of R_3 unless it is negative we can't achieve this criteria when $(R_3 \gg R_2)$ the min. possible value of $70mV/^\circ C$ can be realized.

(c) The $V_{Ref}^{(min)}$ value is limited by the magnitude of V_{EB} for the BJT (0.5 to 0.7V). Therefore it is not possible to precisely generate values which are below V_{EB} .

Q3)

$$+ g_{m1,2} (\gamma_{01} \parallel \gamma_{03}) \quad \text{Stage } = 1$$

$$g_{m,5} (\gamma_{07} \parallel \gamma_{05}) \times 2 \quad \text{(Current mirror)}$$

Poles :-

$$\omega_{p3}^{-1} = (\gamma_{05} \parallel \gamma_{07}) (C_{GD5} + C_{DB5} + C_{GD7} + C_{DB7}).$$

output node.

$$\omega_{p2}^{-1} = (\gamma_{01} \parallel \gamma_{03}) \left[\begin{array}{l} C_{GD1} + C_{DB1} + C_{GD3} + C_{DB3} \\ + C_{GS10} + C_{GD10} \left(1 + \frac{g_{m10}}{g_{m12}} \right) \\ C_{GB5} + C_{GD5} \left(1 + g_{m5} \gamma_{05} \gamma_{07} \right) \end{array} \right]$$

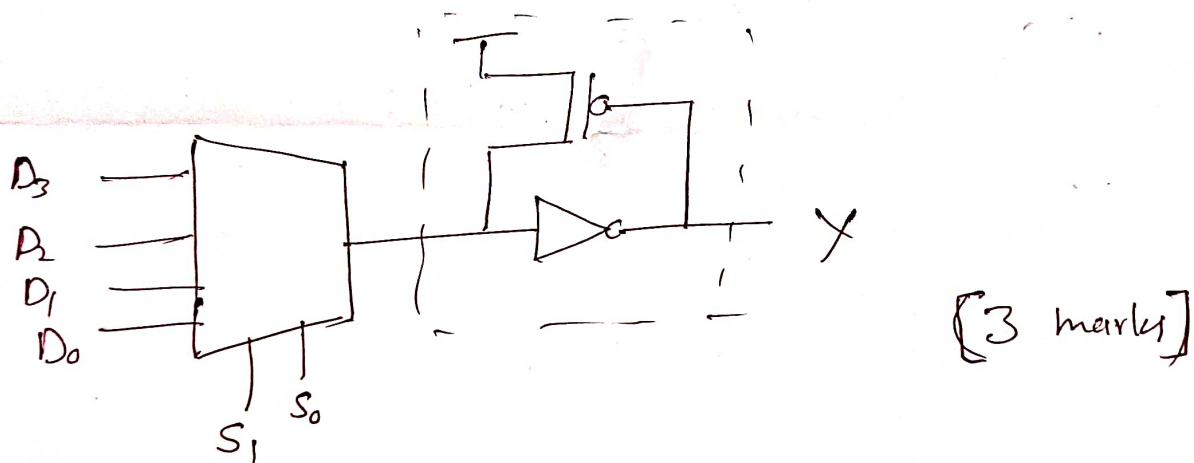
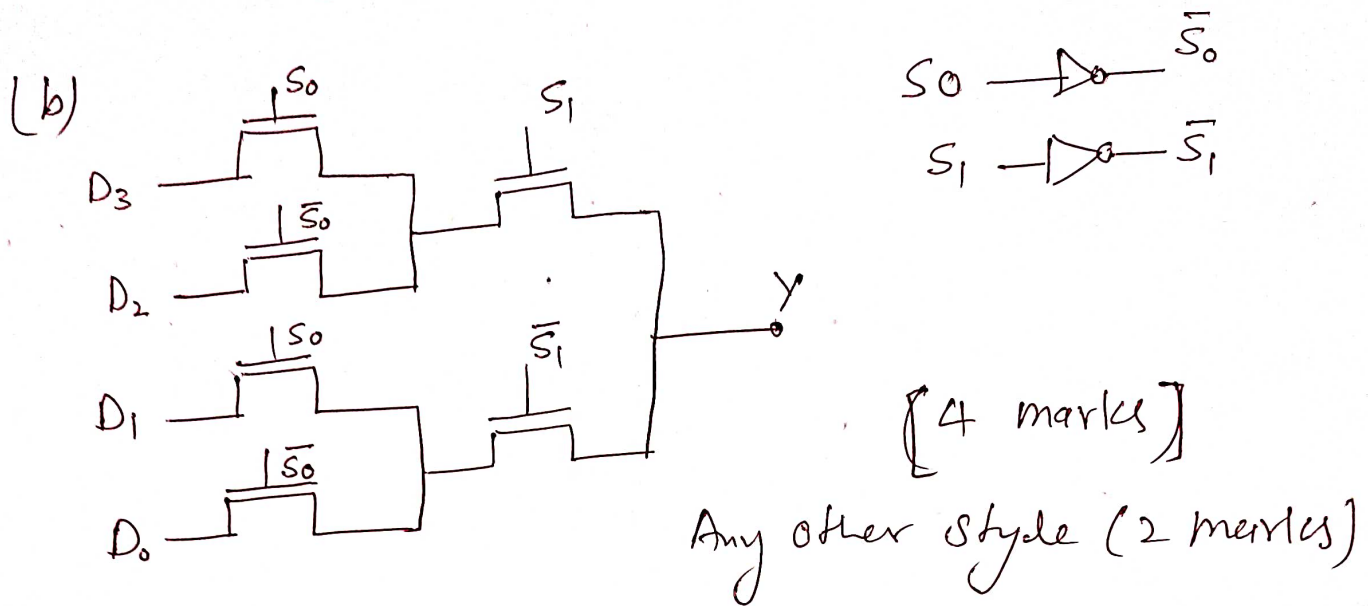
$$\omega_{p1}^{-1} = (\gamma_{09} \parallel \gamma_{011}) \left[\begin{array}{l} C_{GD9} + C_{DB9} + C_{GS11} + C_{DB11} \\ + C_{GS7} + C_{GD7} \left(1 + g_{m7} (\gamma_{05} \parallel \gamma_{07}) \right) \end{array} \right]$$

" Emphasis on Miller components; "

Que 3 Solution (a)

$$d_{HfO_2} = 1.56 \text{ nm} \quad [2.5 \text{ marks}]$$

$$E_{HfO_2} = 3.9 \times 10^3 \text{ V/cm} \quad [2.5 \text{ marks}]$$



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